

The residual, in norm and at every instant

drop-in for Sec. VIII; the norm form is (VIII.3)

0.1 The statement in norm, in one line

Write the recorded rate over the post-onset window as the nominal response, the deviation the model does not carry, and sensor noise,

$$y = \omega_{\text{nom}} + e_{\omega} + n, \quad (\text{P.1})$$

and let $\hat{\omega}$ minimise $\|g - y\|$ over the model family M . The nominal is itself a member of M — it is a hyperbolic cosine with the true amplitude and exponent — so it is one of the candidates the minimiser was choosing among, and therefore

$$\|r\| = \|\hat{\omega} - y\| = \min_{g \in M} \|g - y\| \leq \|\omega_{\text{nom}} - y\| = \|e_{\omega} + n\|. \quad (\text{P.2})$$

That is the whole argument in norm. It uses optimality and nothing else: no linearity, no projection, no expansion, no triangle inequality. It is (VIII.3), and dividing by $\sqrt{\tau_{\text{end}}}$ puts it in RMS form, $\text{RMS}(r) \leq \text{RMS}(e_{\omega}) + \text{RMS}(n)$.

0.2 The same statement at every instant

The pointwise version one would like to write is

$$|r_i| \stackrel{?}{\leq} |e_{\omega,i}| + k\sigma_n, \quad (\text{P.3})$$

and it does not follow from (P.2). A norm inequality constrains a sum of N numbers; about any one of them it says only $|r_i| \leq \|r\|$, which is a bound by the total and not by the local value $|e_{\omega,i}|$.

Nor is (P.3) true, and where it fails is the onset. With C_2 held at the value the rig fixes, the admissible curves form a two-dimensional subspace

$$V = \text{span}\{u, \mathbf{1}\}, \quad u_i = \cosh C_2 \tau_i - 1, \quad (\text{P.4})$$

the least-squares fit is $\hat{\omega} = Py$ with P the orthogonal projector onto V , and P has the closed form

$$P_{ij} = \frac{1}{N} + \frac{\tilde{u}_i \tilde{u}_j}{\sum_k \tilde{u}_k^2}, \quad \tilde{u} = u - \bar{u}. \quad (\text{P.5})$$

The leading $1/N$ alone settles it: a perturbation anywhere in the window shifts the fitted *constant*, and so moves the fitted curve at every sample including the first. Since e_{ω} lives at the far end of the window — its envelope is $E \propto \sinh C_2 \tau$ — its mean displaces the fit at $\tau = 0$ by roughly $0.3 \sup E$ while $e_{\omega}(0)$ itself vanishes. No constant Γ rescues a multiplicative form either: measured on the campaign the ratio of the true bound to E is 2.1 at the end of the window and 27 to 95 at the first sample.

What (P.3) omits is exactly that displacement, and writing it down costs one term. Since $\hat{\omega} = Py$ is an identity — true whether or not the nominal lies in V — substituting (P.1) gives

$$r = (I - P)y = \underbrace{(I - P)\omega_{\text{nom}}}_{\text{misspecification}} + (e_{\omega} + n) - \underbrace{P(e_{\omega} + n)}_{\text{displacement}}, \quad (\text{P.6})$$

exactly, and with no hypothesis at all. Two remarks close it.

The middle group is (P.3)'s right-hand side. Taking absolute values sample by sample returns $|e_{\omega,i}| + |n_i|$, which is what (P.3) already had.

The last group is bounded because it lives in two dimensions. $P(e_{\omega} + n) \in V$, and for $v \in V$ we have $v = Pv$, so $v_i = \langle P_i, v \rangle$ with $\|P_i\|^2 = (P^2)_{ii} = P_{ii}$, whence $|v_i| \leq \sqrt{P_{ii}} \|v\|$ — an identity followed by Cauchy–Schwarz, not an estimate. And $\|P(e_{\omega} + n)\| \leq \|e_{\omega} + n\|$ because P is an orthogonal projection, so (P.2) bounds it too.

Together,

$$|r_i| \leq \left| [(I - P)\omega_{\text{nom}}]_i \right| + |e_{\omega,i}| + |n_i| + \sqrt{P_{ii}} \|e_{\omega} + n\| \quad (\text{P.7})$$

which is (P.3) with two additions: the displacement, the price of the fit being free to move, and the misspecification, the price of the calibrated exponent not being the true one. The first is unavoidable; the second vanishes when the exponents agree.

The noise, separated. Splitting $\|e_\omega + n\| \leq \|e_\omega\| + \|n\|$ and using $|e_{\omega,i}| \leq E_i$, $\|e_\omega\| \leq \|E\|$ from (93) leaves the noise in two places, $|n_i|$ and $\sqrt{P_{ii}}\|n\|$. The second must not be written $\sigma_n\sqrt{N}$: for $n \sim \mathcal{N}(0, \sigma_n^2 I)$ and P symmetric idempotent, $\mathbb{E}\|Pn\|^2 = \sigma_n^2 \text{tr } P = 2\sigma_n^2$ whatever N is, and $\text{Var}[(Pn)_i] = \sigma_n^2 P_{ii}$, so the two together are at most $k\sigma_n(1 + \sqrt{P_{ii}})$ — against a factor $\sqrt{N/2} = 4.5$ at $N = 41$ for the crude route. The form checked below is therefore

$$|r_i| \leq \underbrace{m_i}_{\text{misspec.}} + \underbrace{E_i}_{\text{pointwise}} + \underbrace{\sqrt{P_{ii}}\|E\|}_{\text{displacement}} + k\sigma_n(1 + \sqrt{P_{ii}}), \quad m = |(I - P)\omega_{\text{nom}}|. \quad (\text{P.8})$$

Two Gaussian events enter per sample, so a union bound over the window replaces k by $\Phi^{-1}(1 - \alpha/4N)$: for $N = 53$ that is 3.50 at 95% and 3.90 at 99%, against 1.96 and 2.58 for one sample.

If C_2 is fitted rather than given. There is then no P and (P.6) is unavailable — but (P.2) is untouched, having never used one. Only the pointwise step needs replacing, by the corresponding inequality on the two-dimensional manifold, where the displacement carries a factor $(1 - \|e_\omega + n\|/\varrho)^{-1}$ with ϱ its reach. That is the calibration stage’s case. The per-run fit, which is handed a C_2 , is the linear one throughout.

0.3 The constants, and why they do not grow

Everything in (P.8) is computable before a run is flown: P depends only on C_2 and the sample grid, E only on the geometry box.

$\dot{M}$ N m/s	N	$\text{tr } P$	$\kappa = \max_i \sqrt{P_{ii}}$	$\ E\ /\sup E$	$\kappa\ E\ /\sup E$
0.10	74	2.000	0.352	3.13	1.12
0.20	63	2.000	0.362	3.12	1.15
0.30	57	2.000	0.370	3.10	1.17
0.45	54	2.000	0.376	3.09	1.18
0.65	48	2.000	0.385	3.07	1.21
0.90	44	2.000	0.395	3.02	1.22
1.20	41	2.000	0.404	2.99	1.23

The trace is 2 at every rate and every window length, and (P.5) shows why directly: $\sum_i 1/N = 1$ and $\sum_i \tilde{u}_i^2 / \sum_k \tilde{u}_k^2 = 1$, one unit for each basis direction. That is why the constants do not grow with the record — a two-dimensional projector cannot move much, however many samples it is given. The displacement term is 1.09 to 1.29 times $\sup E$ across all 140 runs, so in uniform form $\max_i |r_i| \lesssim 2.3 \sup E + 1.4 k\sigma_n$.

Equation (P.5) also locates the worst sample: P_{ii} is largest where $|\tilde{u}_i|$ is, which is the last sample of the window in all 140 runs, and only $1/N = 0.013$ to 0.024 of P_{ii} there is the constant — the rest is the amplitude direction, whose cosh has grown by then. It reproduces the numerically formed projector to 10^{-16} .

0.4 The bound tested, and the two things it exposes

Applied to every sample of every run at $k = 3.5$:

$\dot{M}$ N m/s	runs entirely inside σ_{pre}	inside σ_{in}	worst ratio	deployed worst	$\sigma_{\text{pre}} / \sigma_{\text{in}}$ °/s
0.10	20/20	20/20	0.324	0.369	0.087 / 0.831
0.20	20/20	20/20	0.391	0.551	0.044 / 0.800
0.30	18/20	20/20	0.457	0.584	0.101 / 0.823
0.45	19/20	20/20	0.491	0.979	0.127 / 0.707
0.65	18/20	20/20	0.580	0.863	0.122 / 0.856
0.90	15/20	20/20	0.556	1.030	0.111 / 0.749
1.20	13/20	20/20	0.611	1.212	0.173 / 0.773

The bound holds on all 140 runs at every sample, the worst reaching 0.611 of it. For comparison, dropping the displacement term — that is, checking (P.3) itself with E in place of $|e_\omega|$ — holds for 99.5% of samples and 120 of 140 runs entirely. The naive form is nearly right, and the extra term is what closes the last twenty runs.

The noise scale must be measured in the window. Taking σ_n from the pre-onset record leaves 123 of 140 inside, and the shortfall is at the fast end. That is not the bound failing but its hypothesis: before the onset the vehicle is carried by every landing gear, after it pivots on one edge, and the in-window disturbance is 0.71 to $0.86^\circ/\text{s}$ against a pre-onset floor of 0.04 to 0.17 . The honest estimate is the in-window content above 5 Hz, which the fitted model provably cannot produce — $\hat{\omega}$ is smooth on the scale of the window — so whatever is there is disturbance by construction rather than by assumption.

A pinned amplitude adds a term. V is the family the bound is about: amplitude and level free. The shipped estimator frees neither. It takes $C_1 = K\dot{M}$ from the calibration stage, and it forms the level C from the *pre-onset* segment rather than minimising over it on the window, so once the onset is chosen its post-onset curve has no free parameter and $\hat{\omega} \neq Py$. Two terms then survive,

$$|r_i| \leq |\Delta C_1| u_i + |C - b| + m_i + E_i + \sqrt{P_{ii}} \|E\| + k\sigma_n(1 + \sqrt{P_{ii}}),$$

the first being the calibration’s amplitude error and the second the baseline channel of (105)–(106). Their measured sizes are very different. The amplitude term is $|\Delta C_1|/C_1 = 11.3\%$ and it is what puts 26 runs outside the norm form (VIII.3); without it the deployed residual reaches 1.21 of the bound. The baseline term is 0.005 to $0.029^\circ/\text{s}$, under 2% of $\sup E$. One constraint matters and the other does not, and the bound says which.

0.5 What the misspecification term is worth

The term $m = |(I - P)\omega_{\text{nom}}|$ is the whole content of the objection that the fitted curve and the true one do not share a basis. They do not: the nominal carries C_2^* and the span is built on the calibrated C_2 . Writing $C_2 = C_2^*(1 + \varepsilon_2)$, it is the price of conditioning on the calibration stage — which fits C_2 freely, by a per-run nonlinear least squares and a per-configuration median — and it is small, almost linear in ε_2 :

$\dot{M}$ N m/s	$\varepsilon_2 = 1\%$	$\ (I - P)\omega_{\text{nom}}\ _\infty / \sup E$ 2%	5%	10%
0.10	0.005	0.010	0.027	0.059
0.45	0.013	0.027	0.071	0.152
0.90	0.018	0.036	0.093	0.195
1.20	0.020	0.041	0.105	0.221

A 2% exponent error costs 1 to 4% of the envelope and a 5% error costs 3 to 11%, the fast ramp being the more exposed because its window is shorter and the two hyperbolic shapes are less separated there. Read the other way, 98% of a nominal misspecified by 1% still lies inside V — the same near-collinearity of cosh and sinh over this window, 3.4° once the constant is removed, that Sec. VI-E uses elsewhere. A shifted exponent barely leaves the span it is fitted in.

Like the pinned amplitude the term enters additively and is not covered by $\bar{\rho}$, which bounds the physics the model omits and says nothing about the constants the model is given. The empirical check above is unaffected by any of it: it compares a *measured* record, which carries whatever exponent the rig actually has, against the envelope, so the 140/140 already includes the misspecification. What the term supplies is the proof that this had to be so, and a budget line for it.